# SHAHD WALEED ABDELHALEEM

**AI & Data Science | AI Automation**

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## PROFESSIONAL SUMMARY

Computer Science and Artificial Intelligence graduate specializing in AI, with hands-on experience in machine learning, data analytics and AI-powered business automation. Focused on building practical AI and data solutions that automate business processes and support data-driven decisions.

## TECHNICAL SKILLS

**Programming:** Python, SQL, R, Java, C++, PHP, HTML, CSS, JavaScript

**AI & Machine Learning:** Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, Explainable AI, Prompt Engineering, Predictive Analytics

**Data & Analytics:** Pandas, NumPy, Data Cleaning, EDA, Feature Engineering, Statistical Analysis, Data Visualization, Business Intelligence

**Libraries & Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, PySpark, OpenCV, NLTK, Hugging Face, Matplotlib, Seaborn, Plotly, Dash, Django

**Automation & Integration:** n8n, AI Agents, Google Gemini, OpenAI API, MCP, REST APIs, Webhooks, Airtable, Google Sheets, Telegram, WhatsApp API, Redis

**Tools:** Git, GitHub, Jupyter, Google Colab, Kaggle, Postman, MySQL, Power BI, Figma, UML, ERD

## PROJECTS

### AI-Powered Multi-Agent Business Automation System (Capstone)

- Designed an AI system in n8n that automates CRM, HR recruitment, inventory management, social media publishing and monitoring, connecting Google Gemini, Airtable, Google Sheets, Telegram and webhooks in one platform.
- Applied AI agents, prompt engineering and MCP to link AI capabilities with real business processes instead of isolated models.

  **Tech Stack:** *n8n, Google Gemini, Airtable, Google Sheets, Telegram, REST APIs, Webhooks*

### AI Agent Systems (n8n)

- Built a WhatsApp appointment-booking agent for a hospital with Redis slot locking, a multi-agent SEO content system, and an inventory demand-forecasting workflow that generates purchase orders.

  **Tech Stack:** *n8n, OpenAI API, Google Gemini, WhatsApp API, Redis, Google Sheets*

### Mental Health Treatment Prediction (PySpark)

- Built a big-data pipeline with Spark SQL analysis, feature engineering and a comparison of Logistic Regression, Random Forest and Gradient Boosting to predict treatment-seeking.

  **Tech Stack:** *Python, PySpark, Spark SQL*

### Deep Learning & Computer Vision

- Developed a pneumonia detector for chest X-rays using DenseNet121 transfer learning, plus autoencoder anomaly detection and facial-expression recognition experiments.

  **Tech Stack:** *TensorFlow, Keras, PyTorch, OpenCV*

### Smart Tourism System

- Designed the Figma UI and built a PHP/MySQL platform for trip planning, map, bookings and recommendations, with UML and ERD design.

  **Tech Stack:** *PHP, MySQL, Figma, UML, ERD*

### Business Intelligence & Dashboards

- Created interactive Power BI dashboards (HR analytics, COVID-19 trade) and a Plotly/Dash attrition dashboard for KPI monitoring and decision-making.
  **Tech Stack:** *Power BI, Plotly, Dash*

### NLP & Information Retrieval

- Built a TF-IDF search engine with Boolean and phrase queries; team project reproducing an LLM-based SMS spam classification study.
  **Tech Stack:** *NLTK, Hugging Face Transformers*

### Data Science & Machine Learning

- Applied cleaning, EDA, feature engineering and modeling (classification, regression, clustering) to loan, housing, stroke, sales and employee datasets.
  **Tech Stack:** *Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn*

### Django Web Application

- Developed a client-management site with Django models, templates, admin panel and SQLite.
  **Tech Stack:** *Django, Python, SQLite*

## EDUCATION

**BSc Computer Science and Artificial Intelligence**
Specialization: Artificial Intelligence | Beni Suef National University, Egypt | Graduation Year: 2026

## CERTIFICATIONS & COURSES

- Machine Learning Specialization — Kaggle
- Artificial Intelligence Fundamentals — IBM
- Full Stack Web Development — NTI
- Data Science Foundations — Coursera

## RELEVANT COURSEWORK

Mathematics, Statistics, Probability, Data Science, Machine Learning, Data Visualization, Database Systems, Big Data Analytics, Software Engineering, Data Structures and Algorithms